

# Evaluation of a guardian plant system to suppress *Frankliniella occidentalis* (Thysanoptera: Thripidae) in greenhouse ornamentals

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## Abstract

**BACKGROUND:** Western flower thrips (WFT), *Frankliniella occidentalis* (Pergande), is an economically damaging pest of greenhouse ornamentals. A 'guardian plant system' (GPS) that targeted WFT was evaluated under controlled and commercial greenhouse conditions. This system used mycotized millet grains with the entomopathogenic fungus *Beauveria bassiana* (Balsamo-Crivelli) Vuillemin applied to soil of potted marigolds, *Tagetes patula* (L.), combined with the foliar-dwelling predatory mite *Neoseiulus* (= *Amblyseius*) *cucumeris* (Oudemans) in slow-release sachets under controlled greenhouse conditions, and with the addition of a pheromone lure under commercial settings.

**RESULTS:** Significantly fewer WFT and less foliar damage on GPS was observed over the 10 and 12 weeks experimental periods compared to the untreated controls. Predatory mites were maintained up to 10 weeks with one release under controlled greenhouse conditions and 12 weeks with two releases in commercial greenhouses. In commercial greenhouses, greater numbers of WFT were found on marigolds than on crop plants within 1 m of the system. Fungal granules persisted for 12 weeks up to  $2.5 \times 10^5$  CFU g<sup>-1</sup> in the GPS soil.

**CONCLUSION:** The use of biological control agents to suppress WFT within a GPS could be a useful IPM strategy for greenhouse production. The marigold GPS attracted WFT which were suppressed primarily through predation by foliar-dwelling predatory mites and to a lesser extent, infection from conidia produced by the granular fungal formulation in the soil. Further investigations into system deployment and fungal granular application rates and new fungal formulations are suggested to improve system efficacy.

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Supporting information may be found in the online version of this article.

**Keywords:** western flower thrips; trap crop; biological control; biopesticide; *Beauveria bassiana*; integrated pest management; *Neoseiulus cucumeris*

## 1 INTRODUCTION

Western flower thrips (WFT), *Frankliniella occidentalis* (Pergande) [Thysanoptera: Thripidae], are one of the most important pests of crops grown under protected culture worldwide.<sup>1–3</sup> Management with synthetic chemical pesticides is challenging due to widespread resistance.<sup>4–6</sup> As a result, there is a need for alternative strategies that growers can adopt to reduce the need for repeat chemical pesticide applications.<sup>7</sup> New management strategies must be reliable, easy to understand and deploy and be cost-effective for growers to adopt them.<sup>8,9</sup>

Integrated pest management (IPM) uses an array of practices, including population monitoring/scouting and biological and cultural controls to reduce pest problems and grower reliance on synthetic chemical pesticides.<sup>7,9–11</sup> The entomopathogenic fungus *Beauveria bassiana* (Balsamo) Vuillemin [Hypocreales: Cordycipitaceae] and the predatory mite *Neoseiulus* (= *Amblyseius*)

*cucumeris* (Oudemans) [Acari: Phytoseiidae] are biological control agents commonly used against WFT.<sup>12–18</sup> *Beauveria bassiana* formulations are typically applied to either the foliage or soil of potted plants whereas *N. cucumeris* is applied to the foliage loosely or in hanging sachets/bags. Combining compatible biological

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control agents can enhance IPM efficacy. The effectiveness of combined releases of various predatory mite species and entomopathogens against WFT has been demonstrated.<sup>19–23</sup> This strategy could be useful by targeting pests that have life stages that occur both on plant foliage and within the soil as for WFT.<sup>3</sup>

Strategies using secondary plants, which are not grown for profit, are integral components of IPM. These divert pests away from the main crop using a more enticing food source.<sup>24–27</sup> Plant-mediated IPM systems (indicator, trap, banker, habitat, guardian plants) utilize plants in combination with other tactics to manage arthropod pests in greenhouse-grown crops.<sup>11</sup> Indicator or sentinel plants are species known to be attractive to pests to facilitate early detection.<sup>28</sup> Trap plants lure the pest from the crop and can then be removed, or offer a site for targeted suppression, such as releasing natural enemies or applying pesticides.<sup>29–31</sup> Habitat, insectary or banker plants are grown to provide a continual food source and suitable environment for naturally occurring or commercially produced biological control agents.<sup>32–35</sup> Guardian plants combine all the functions of the previously described systems at once.<sup>13,36–38</sup>

Pest detection for early intervention is a cornerstone of effective IPM for WFT and usually relies on use of yellow or blue sticky traps to attract and capture winged adults.<sup>39–41</sup> However, indicator/trap plants can also serve this function. For example, flowering, yellow chrysanthemums, *Dendranthema grandiflora* (Tzvelev) and marigolds, *Tagetes patula* (L.) [Asterales: Asteraceae] have been successfully used as trap plants for WFT in greenhouse ornamentals.<sup>29,42,43</sup> Chemical/pheromone attractants can be added to enhance catches. They have been used with sticky cards<sup>44</sup> and combined with trap plants in field and orchard crops.<sup>45,46</sup> Although semiochemical lures have been paired with sticky cards in greenhouse ornamentals for WFT,<sup>47</sup> pairings with plant-mediated IPM systems are understudied.

The drawback of a system that is highly attractive to WFT is the risk of it becoming a pest source which could jeopardize the health of surrounding crops. To reduce this risk, biological control agents could be applied at the start of the system for early establishment to provide suppression at the onset of infestation. Herein we report on tests of the efficacy of a guardian plant system (GPS) to manage WFT under controlled and commercial greenhouse settings. This system used a flowering marigold to attract WFT out of the crop. A granular formulation of mycotized millet grains that contained *B. bassiana* was incorporated into the soil to target pupating WFT. Predatory mites, *N. cucumeris*, in slow-release sachets were released to attack larvae in the above ground part of the plant that attracted WFT adults. In the commercial greenhouse trials a pheromone lure (Neryl (S)-2-methyl butanoate) was incorporated into the system to enhance GPS attractiveness to WFT.

## 2 MATERIALS & METHODS

### 2.1 Fungal isolates and granular production

A solid-substrate culture method was used to produce mycotized millet grains.<sup>16</sup> Two isolates of *B. bassiana* were selected for these trials: ERL1170 (ARS7060) and GHA, from the commercial product BotaniGard®WP (LAM International, Butte, MT, USA). Isolates are held at the Entomology Research Laboratory Worldwide Collection of Entomopathogenic Fungi at the University of Vermont, Burlington, Vermont, USA. The experimental isolate was selected because it displayed high virulence against WFT in previous laboratory bioassays.<sup>16,48</sup> Prior to use in experiments, viability was determined by examining conidia germination.<sup>16</sup> Plates of SDAY were inoculated with 0.1 mL of a conidial suspension from the

granules and held at 25 °C for 24 h. From each plate, 400 conidia were inspected and those with germination tubes longer than their width indicated acceptable germination.<sup>49</sup> Granular batches with germination rates >98% were used in experiments.

### 2.2 Controlled greenhouse experiment

Six treatments were tested on approximately 17 cm-tall flowering marigolds (variety 'Hero Yellow') grown from seed (Ball Horticultural Company, West Chicago, IL, USA) in Metro-Mix®360 (Sun Gro Horticulture, Agawam, MA, USA) within 12.7 cm diameter, round, plastic azalea pots (Dillen; The HC Companies, Twinsburg, OH, USA): 1 and 2: *B. bassiana* (ERL1170) with and without predatory mites; 3 and 4: *B. bassiana* (GHA) with and without predatory mites; 5: mites with no fungi; 6: untreated control (no mites or fungi). Predatory mite, *N. cucumeris*, sachets were produced by Applied Bio-nomics Ltd., (North Saanich, BC, CAN). Sachets contained *Tyrophagus* sp. mites in a bran substrate as a supplementary food source for the predatory mites. Sachets were hung from an upper branch of the marigold on the day fungi were added to the potting mix.

Trials were conducted within a temperature-controlled glass greenhouse (5.4 × 6.1 m) held at ~23–25 °C in cages (0.75 × 0.75 × 0.6 m) made from thrips-proof screening. Each cage contained four plants and was randomly assigned a treatment. Each treatment was replicated two times within the greenhouse. For each fungal treatment, 1 × 10<sup>8</sup> conidia / g (4 g total) of mycotized millet grains were applied once at the start of the experiment to the potting soil surface of each pot (126.67 cm<sup>2</sup>) and mixed gently into the upper 3 cm. One-week post-application, five 24-h old laboratory-reared mated females were released onto each plant (20 WFT/cage). The total number of WFT larvae and adults for each plant was counted biweekly for 10 weeks by gently tapping the entire plant canopy over a 28 × 22 cm piece of laminated white paper. Dislodged WFT were returned to the same plant canopy after sampling. Foliar damage from thrips feeding (silvering or stippling on the leaf surface, leaf curling, stunting and distortion) on each plant was assessed visually using the following rating scale: 0: no damage; 1: <10% of the foliage damaged; 2: 10–25% of the foliage damaged; 3: 26–50% of the foliage damaged; 4: 51–75% of the foliage damaged; and 5: >75% of the foliage damaged.

After six (mid-experiment) and 10 weeks (end of experiment), four blossoms per cage (one per marigold) were collected, placed in 75% ethanol and dissected microscopically to count the number of thrips and predatory mites on plants remaining after tapping onto the laminated paper. The entire experiment was repeated twice (one time per year) over two consecutive years. Newly produced mycotized millet grains and predatory mite sachets were used for each experiment.

### 2.3 Commercial greenhouse experiments

Marigold GPS were tested in greenhouses at six sites (five in Vermont and one in New Hampshire, USA) over three consecutive years spanning late February through early June, depending on the production cycle of the operation. Trials were conducted in typical greenhouses found in the Northeastern, USA (gutter-connected, single-gable or free-standing Quonset style). Greenhouse sizes ranged from 186–743 m<sup>2</sup> and averaged 551 m<sup>2</sup>. Crops grown within the greenhouses were mixtures of ornamental bedding plants, herbs and vegetable starts and differed among sites, greenhouses and years. All sites practiced IPM and used biological control and/or spot applications of pesticides for pest outbreaks when necessary. Marigold GPS did not receive additional releases

of natural enemies or any pesticide applications during the experiment. Natural infestations of WFT were relied upon for this experiment.

In each greenhouse, four treatments were tested on approximately 17 cm-tall flowering marigolds in 17.78 cm diameter, round plastic azalea pots in the same growing mix as described above: 1: *B. bassiana* (ERL1170/ARS7060) with a predatory mite sachet and a thrips lure (ThriPher, Biobest, Leamington, ON, CAN); 2: *B. bassiana* (GHA) with predatory mites and the thrips lure; 3: thrips lure only; 4: untreated control (no fungi, mites or lure). For the fungal treatments,  $1 \times 10^8$  conidia / g (13.2 g total) of fungal granules were applied over the 248.28 cm<sup>2</sup> area of potting mix in the pot. Predatory mites in sachets were released two times, at the beginning of the experiment and after 6 weeks per supplier guidelines. Lures were placed once on plants at the beginning of the experiment. Each treatment was replicated three times within each greenhouse. The GPS were placed equidistant within each greenhouse ranging from 1/16 m<sup>2</sup> to 1/62 m<sup>2</sup> with an average rate of 1/25 m<sup>2</sup>. Marigolds were checked for WFT and foliar damage bi-weekly for 12 weeks as described for the cage trial. In addition, three crop plants that were adjacent to the marigold GPS were randomly selected on each observation date and inspected for WFT. Similar to the controlled greenhouse cage trials, two blossoms were collected from each marigold and placed in 75% ethanol to assess the number of WFT and predatory mites over time. Collected thrips were periodically identified to ensure WFT were the dominant species occurring within the greenhouses.

The fungal concentration in the marigold GPS potting mix was determined after inoculation and at the end of the experiments. A 20 g sample was taken from each pot and mixed with a 0.02% organosilicone surfactant (Silwet L-77<sup>®</sup>; Helena Agri-Enterprises, Collierville, TN, USA) solution to make a 50-ml soil suspension. Diluted soil suspensions (100 µL) were spread on Petri dishes containing a selective medium for *B. bassiana*<sup>50</sup> and held at  $\pm 25^\circ\text{C}$  for 3–5 days, after which the number of colony forming units (CFU) was determined per gram of potting soil.

## 2.4 Data analysis

Data from all experiments on the number of WFT (adults and/or larvae) dislodged from plants, the mean number of predatory mites and WFT per blossom, WFT foliar damage rating, the

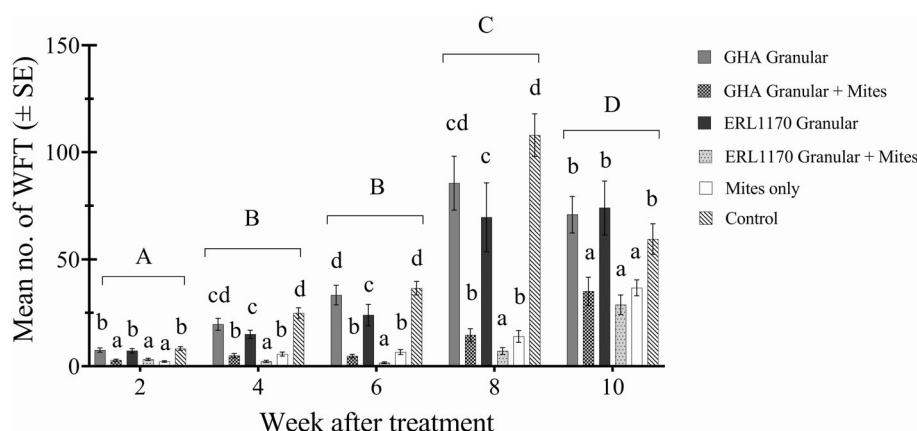
number of fungal CFU, ratio of larvae and mean number of WFT on crop plants associated with marigold GPS were analyzed using a general linear model with repeated measures and/or univariate procedures, followed by Tukey's honestly significant difference (HSD) tests ( $\alpha = 0.05$ ). For the commercial greenhouse trials, there were different crops grown and various management strategies used, therefore, we considered the greenhouses to be independent across year and site. Because we relied upon natural infestations of WFT, we selected datasets from 15 greenhouses based on their overall mean infestation level of >10 WFT/marigold/greenhouse at the end of the 12 weeks experimental period. These were selected to observe the effect of the system on attracted WFT in greenhouses exhibiting a 'moderate' infestation level of WFT. To determine the attractiveness of marigolds to WFT, the total number of WFT per marigold were compared to the mean number of WFT from the three crop plants per marigold on the first observation date. T-tests were used to determine significant differences between the plant types after the univariate analysis. To observe overall GPS treatment effects, analyses were conducted from week 6 through week 12 after WFT populations increased on GPS. The number of WFT on marigolds in both the controlled and commercial greenhouses were log transformed to correct for normality and sphericity. All other data were not transformed, and untransformed data are reported for all results. Analyses were conducted using IBM-SPSS v.26 (IBM, Armonk, NY, USA).

## 3 RESULTS

### 3.1 Controlled greenhouse experiment

#### 3.1.1 WFT and foliar damage on marigolds

Significant differences in the mean number of WFT were observed among the fungal granular and predatory mite treatments over the 10 weeks experimental period ( $F_{20,232} = 10.03$ ;  $P < 0.001$ ). The mean number of WFT increased over time on the marigolds ( $F_{4,232} = 377.31$ ;  $P < 0.001$ ) and there were consistently significantly fewer WFT on marigold GPS that contained predatory mites both with and without fungal granules than marigolds with fungus only and the untreated control. ERL1170 without mites had significantly fewer WFT compared to the control within weeks 4, 6 and 8, whereas GHA without mites did not within any week (Fig. 1).



**Figure 1.** Mean number of *Frankliniella occidentalis* (WFT) adults and larvae dislodged from marigolds treated with granular fungal formulations of *Beauveria bassiana* experimental isolate ERL1170 or commercial isolate GHA with or without *Neoseiulus cucumeris* predatory mites within slow-release sachets over 10 weeks under controlled greenhouse conditions. Means with the same upper-case letters among weeks or the same lower-case letters among treatments within a week are not significantly different (Tukey's HSD,  $P > 0.05$ ).

Because predatory mites primarily feed on WFT larvae rather than adults, an assessment of the proportion of these life stages in the foliar samples could further demonstrate the direct impact of this predator as a biological control agent. Significant differences were observed in the percent of WFT larvae on marigolds among treatments over time ( $F_{20,288} = 2.0$ ;  $P = 0.007$ ) (Table 1). On marigolds treated with the predatory mites, the average proportion of larvae ranged from 7–41% whereas on marigolds without predatory mites, ranged from 26–74%. In addition, significant differences were detected in the proportion of WFT larvae over time ( $F_{4,288} = 6.81$ ;  $P < 0.001$ ). In most sample weeks, marigolds with predatory mites had significantly lower percentage of larvae compared to those without mites ( $P < 0.001$ ). After 10 weeks, although there were fewer larvae on plants with predatory mites than plants without mites, the differences were not significant.

After 8 weeks, the untreated control marigolds had a mean of  $107.94 \pm 9.96$  WFT/plant resulting in rapid deterioration in foliar appearance which led to reduced WFT numbers at week 10. Damage from WFT feeding on marigold foliage significantly increased over time during the experimental period ( $F_{20,288} = 359.48$ ;  $P < 0.001$ ) and there were significant differences among treatments (Table 2). After 8 weeks, the untreated control had significantly greater numbers of WFT than all other treatments. When

the mean number of WFT among treatments were assessed for the 8-week-period (omitting week 10) prior to plant deterioration, when WFT were at their highest, significant differences were still observed ( $F_{15,174} = 6.87$ ;  $P < 0.001$ ).

### 3.1.2 WFT and predatory mites within marigold blossoms

There were significant differences in the mean number of remaining WFT per blossom among treatments over the experiment duration ( $F_{5,12} = 3.32$ ;  $P = 0.041$ ). Of the total WFT observed in blossoms, 4% were adults, indicating larvae (96%) were the primary stage remaining in the blossoms. After 6 weeks there were significant differences in remaining WFT among treatments ( $F_{5,18} = 4.75$ ;  $P = 0.006$ ), but not after 10 weeks ( $F_{5,18} = 0.26$ ;  $P = 0.930$ ), when the experiment ended (Table 3). In general, marigold GPS treated with predatory mites, with or without fungal granules, had significantly fewer WFT per blossom than the untreated control. The marigolds that received mite sachets had a mean of  $1.60 \pm 0.44$  mites per blossom after 6 weeks and  $0.95 \pm 0.27$  after 10 weeks. Differences in the mean number of mites per blossom among treatments with mite sachets were not significant during the experiment ( $F_{2,6} = 0.94$ ;  $P = 0.442$ ). However, there were significantly fewer mites per blossom at 10 weeks than 6 weeks ( $F_{1,6} = 12.36$ ,  $P = 0.013$ ).

**Table 1.** *Frankliniella occidentalis* (WFT) larvae (%) on marigolds untreated or treated with granular fungal formulations of *Beauveria bassiana* experimental isolate ERL1170 and commercial isolate GHA with or without *Neoseiulus cucumeris* predatory mites over 10 weeks post treatment under controlled greenhouse conditions

| Treatment               | % Larvae |            |           |           |           |
|-------------------------|----------|------------|-----------|-----------|-----------|
|                         | 2 weeks  | 4 weeks    | 6 weeks   | 8 weeks   | 10 weeks  |
| Control (marigold only) | 74 ± 4 b | 31 ± 2 bc  | 44 ± 4 ab | 54 ± 6 c  | 62 ± 31 a |
| ERL1170                 | 61 ± 7 b | 26 ± 5 abc | 58 ± 6 b  | 46 ± 3 bc | 30 ± 3 a  |
| GHA                     | 55 ± 5 b | 38 ± 5 c   | 52 ± 4 b  | 50 ± 5 bc | 66 ± 24 a |
| Mites only              | 10 ± 6 a | 18 ± 5 ab  | 20 ± 6 a  | 27 ± 4 a  | 34 ± 3 a  |
| ERL1170 + mites         | 17 ± 6 a | 9 ± 4 a    | 26 ± 10 a | 22 ± 5 a  | 41 ± 5 a  |
| GHA + mites             | 7 ± 4 a  | 11 ± 4 a   | 25 ± 7 a  | 36 ± 6 ab | 33 ± 3 a  |
| $F_{5,84}$              | 37.60    | 7.18       | 6.71      | 10.56     | 1.10      |
| $P$                     | < 0.001  | < 0.001    | < 0.001   | < 0.001   | 0.369     |

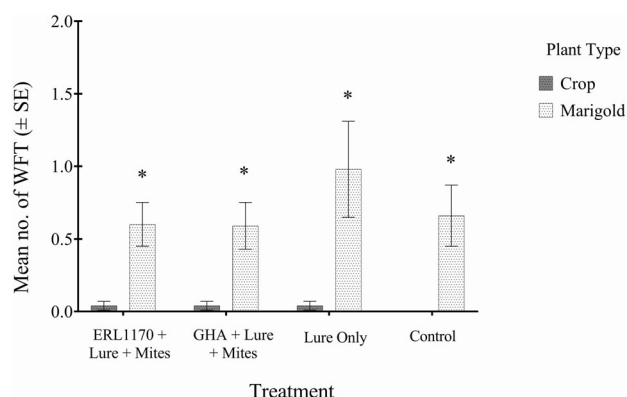
Note: Means within columns with the same lowercase letters are not significantly different (Tukey's HSD,  $P > 0.05$ ).

**Table 2.** Mean foliar damage rating from marigolds untreated or treated with granular fungal formulations of *Beauveria bassiana* experimental isolate ERL1170 and commercial isolate GHA with or without *Neoseiulus cucumeris* predatory mites over 10 weeks post treatment under controlled greenhouse conditions

| Treatment               | Mean foliar damage rating* |                |               |               |               |
|-------------------------|----------------------------|----------------|---------------|---------------|---------------|
|                         | 2 weeks                    | 4 weeks        | 6 weeks       | 8 weeks       | 10 weeks      |
| Control (marigold only) | 1.00 ± 0.00 a              | 1.50 ± 0.13 a  | 2.81 ± 0.14 a | 4.50 ± 0.13 a | 5.00 ± 0.00 a |
| ERL1170                 | 0.75 ± 0.11 ab             | 0.94 ± 0.06 b  | 1.00 ± 0.00 d | 1.25 ± 0.11 d | 2.25 ± 0.28 c |
| GHA                     | 1.00 ± 0.00 a              | 1.50 ± 0.13 a  | 2.25 ± 0.21 b | 4.00 ± 0.26 b | 4.75 ± 0.11 a |
| Mites only              | 0.63 ± 0.13 b              | 0.69 ± 0.12 c  | 1.00 ± 0.00 d | 1.25 ± 0.11 d | 2.25 ± 0.28 c |
| ERL1170 + mites         | 0.94 ± 0.06 a              | 1.50 ± 0.13 a  | 1.75 ± 1.00 c | 3.25 ± 0.34 c | 3.50 ± 0.39 b |
| GHA + mites             | 0.63 ± 0.13 b              | 0.75 ± 0.11 bc | 0.21 ± 0.00 d | 1.00 ± 0.00 d | 1.50 ± 0.22 d |
| $F_{5,84}$              | 7.30                       | 44.51          | 116.81        | 545.53        | 93.43         |
| $P$                     | < 0.001                    | < 0.001        | < 0.001       | < 0.001       | < 0.001       |

\*Foliar damage rating scale: 0: no damage; 1: <10%; 2: 10–25%; 3: 26–50%; 4: 51–75%; and 5: >75%. Means within columns with the same lowercase letters are not significantly different (Tukey's HSD,  $P > 0.05$ ).





**Figure 2.** Mean number of *Frankliniella occidentalis* (WFT) adults and larvae dislodged from marigold GPS untreated or treated with granular fungal formulations of *Beauveria bassiana* experimental isolate ERL1170 or commercial isolate GHA, *Neoseiulus cucumeris* predatory mites within slow-release sachets and a pheromone lure and marigolds with and without the lure and associated crop plants on the first observation date (2 weeks after initial treatment) under commercial greenhouse conditions. Means marked with an \* within treatment group are significantly different ( $P < 0.05$ ).

**Table 3.** Mean number of *Frankliniella occidentalis* (WFT) per blossom from marigolds untreated or treated with granular fungal formulations of *Beauveria bassiana* experimental isolate ERL1170 and commercial isolate GHA with or without *Neoseiulus cucumeris* predatory mites within slow-release sachets after 6 and 10 weeks post treatment under controlled greenhouse conditions

| Treatment               | Mean number of WFT/blossom |                |
|-------------------------|----------------------------|----------------|
|                         | 6 weeks                    | 10 weeks       |
| Control (marigold only) | 21.94 ± 5.89 b             | 11.88 ± 6.69 a |
| ERL1170                 | 12.06 ± 5.89 ab            | 10.63 ± 3.11 a |
| GHA                     | 17.69 ± 3.68 ab            | 14.13 ± 7.72 a |
| Mites only              | 3.25 ± 1.18 a              | 11.56 ± 1.76 a |
| ERL1170 + mites         | 2.06 ± 5.89 a              | 7.94 ± 4.21 a  |
| GHA + mites             | 4.19 ± 1.12 a              | 7.69 ± 2.70 a  |
| $F_{5,24}$              | 4.73                       | 0.26           |
| $P$                     | 0.006                      | 0.930          |

Note: Means within columns with the same lowercase letters are not significantly different (Tukey's HSD,  $P > 0.05$ ).

### 3.2 Commercial greenhouse experiment

#### 3.2.1 Initial marigold attractiveness to WFT

On the first observation date (2 weeks after treatment and placement in greenhouses) significant differences in the mean number of WFT were observed on plants among treatments ( $F_{7,346} = 5.74$ ,  $P < 0.001$ ). WFT were found on both marigolds and crop plants. Crop plants that were within 1 m of marigolds with lures had WFT detected on them whereas no WFT were observed on crop plants next to marigolds without lures. The total number of WFT on marigold GPS ranged from 0–13 WFT/plant whereas on crop plants numbers ranged from 0–2 WFT/plant. There was no interaction effect between plant type (marigold vs. crop plants) and the treatments ( $F_{3,346} = 0.63$ ,  $P = 0.596$ ). In general, there were no significant differences in the number of WFT on plants among treatments ( $F_{3,346} = 0.687$ ,  $P = 0.561$ ) but the number of WFT on the marigolds was significantly greater than on the surrounding

crop plants ( $F_{1,346} = 36.25$ ,  $P < 0.001$ ). This suggests the marigold GPS is effective at attracting WFT from the crops (Fig. 2).

#### 3.2.2 WFT abundance and foliar damage on marigolds

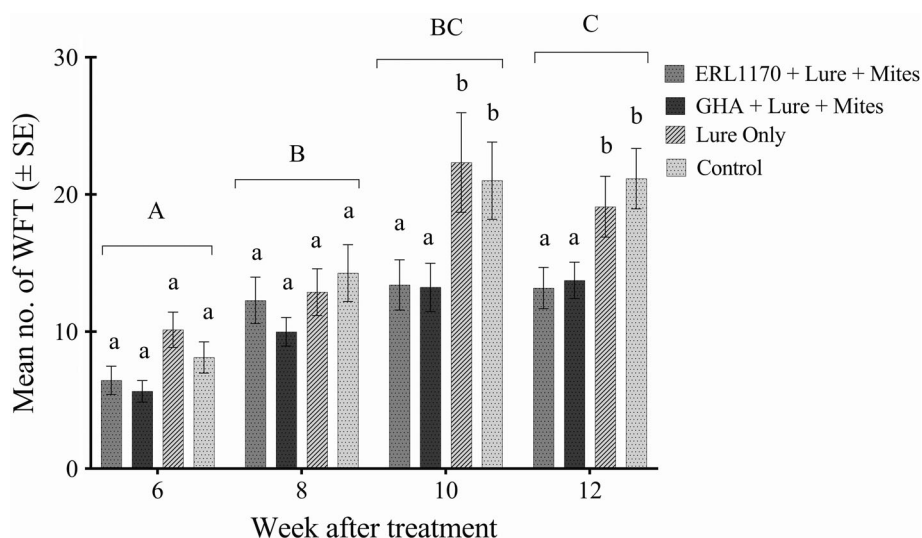
Significant differences in the mean number of WFT were observed among treatments from weeks 6–12 when tested in commercial greenhouses ( $F_{9,210} = 2.65$ ;  $P = 0.006$ ). WFT populations increased significantly over time on the marigolds in all treatments ( $F_{3,210} = 36.27$ ;  $P < 0.001$ ). Throughout the experiment, there was generally no difference within each observation period in WFT abundance on marigold GPS containing either the experimental or commercial *B. bassiana* isolate combined with predatory mites and the pheromone lure. There were significantly fewer WFT on the marigold GPS that combined a fungus + lure + predatory mites than on the untreated controls (marigolds with or without a lure). Differences in WFT numbers on marigolds with only a lure or no added treatments were not significant within a sample week, suggesting that marigolds were naturally attractive, even without a pheromone lure (Fig. 3).

When the ratio of WFT larvae to adults was evaluated on marigold GPS, no significant differences among treatments over time was observed in the test greenhouses ( $F_{9,309} = 1.57$ ;  $P = 0.122$ ) except on week 12. After 12 weeks, the average proportion of larvae for a given treatment did not exceed 29% and that was observed on the untreated control marigold (Table 4). There were significant differences in the proportion of larvae among weeks ( $F_{3,309} = 18.83$ ;  $P < 0.001$ ). The proportion of larvae increased significantly from weeks 6–10 ( $P \leq 0.027$ ), decreasing after 12 weeks to a level that was not significant from weeks 8 and 10 ( $P \geq 0.095$ ). There were significant differences among treatments when they were averaged over the experimental period ( $F_{3,103} = 5.78$ ;  $P = 0.001$ ). In general, the marigolds with lures + mites + fungi ( $P = 0.894$ ) and marigolds without predatory mites ( $P = 0.975$ ) showed no difference in the proportion of larvae. The lure only marigold had a significantly greater proportion of larvae than the two marigold treatments with lures + mites + fungi ( $P < 0.017$ ) whereas the control was only significantly different from the marigold + GHA + lure + mite treatment ( $P = 0.008$ ).

There were significant differences in visual damage on marigold GPS after 12 weeks ( $F_{3,149} = 2.09$ ;  $P = 0.002$ ). The mean level of foliar damage on the treated marigolds after 12 weeks was: 1.18 ± 0.13 (ERL1170 + lure + mites), 1.11 ± 0.10 (GHA + lure + mites), 1.96 ± 0.18 (lure only) and 1.96 ± 0.18 (untreated control), equal to <10% for the GPS treatments and 10–25% for untreated marigolds. The main effect difference among treatments was significant ( $F_{3,149} = 27.97$ ;  $P < 0.001$ ) and damage was significantly less on marigold GPS with mites and granular fungi than those without treatments ( $P < 0.001$ ). There were no differences in the damage rating between the two GPS with fungi + mite + lure treatments ( $P = 0.954$ ) nor were differences detected between marigold plants with and without the lure ( $P = 1.000$ ).

#### 3.2.3 WFT abundance and foliar damage on crop plants

A total of 129 different plant types were inspected that were located within 1 m of the marigold GPS, including ornamentals (86%), herbs (11%) and vegetable starts (3%) (Supplementary information Table 1). The total number of WFT observed on an individual crop plant ranged from 0 to 18 during weeks 6–12. There was no significant interaction effect between the mean number of WFT on crop plants among marigold GPS treatments and observation weeks ( $F_{9,439} = 0.482$ ;  $P = 0.887$ ). In general, the



**Figure 3.** Mean number of *Frankliniella occidentalis* (WFT) adults and larvae dislodged from marigold GPS untreated or treated with granular fungal formulations of *Beauveria bassiana* experimental isolate ERL1170 or commercial isolate GHA, *Neoseiulus cucumeris* predatory mites within slow-release sachets and a pheromone lure and marigolds with and without the lure 6–12 weeks after initial treatment under commercial greenhouse conditions. Means with the same upper-case letters among weeks or the same lower-case letter among treatments within a week are not significantly different (Tukey's HSD,  $P > 0.05$ ).

**Table 4.** *Frankliniella occidentalis* (WFT) larvae (%) on marigold GPS untreated or treated with granular fungal formulations of *Beauveria bassiana* experimental isolate ERL1170 or commercial isolate GHA in combination with *Neoseiulus cucumeris* predatory mites within slow-release sachets and a pheromone lure and marigolds with and without the lure from weeks 6–12 after initial treatment under commercial greenhouse conditions

| Treatment               | % Larvae |          |          |          |
|-------------------------|----------|----------|----------|----------|
|                         | 6 weeks  | 8 weeks  | 10 weeks | 12 weeks |
| Control (marigold only) | 9 ± 2 a  | 19 ± 4 a | 28 ± 4 a | 17 ± 2 a |
| Mites only              | 11 ± 3 a | 13 ± 2 a | 25 ± 4 a | 29 ± 4 b |
| ERL1170 + mites + fungi | 8 ± 3 a  | 11 ± 3 a | 18 ± 4 a | 13 ± 3 a |
| GHA + mites + fungi     | 4 ± 2 a  | 10 ± 2 a | 18 ± 3 a | 13 ± 2 a |
| $F_{3,120}$             | 1.59     | 2.33     | 2.58     | 5.85     |
| $P$                     | 0.195    | 0.078    | 0.057    | 0.001    |

Note: Means within columns with the same lowercase letters are not significantly different (Tukey's HSD,  $P > 0.05$ ).

**Table 5.** Mean number of *Frankliniella occidentalis* (WFT) on crop plants within 1 m of the marigold GPS untreated or treated with granular fungal formulations of *Beauveria bassiana* experimental isolate ERL1170 or commercial isolate GHA in combination with *Neoseiulus cucumeris* predatory mites within slow-release sachets and a pheromone lure and marigolds with and without the lure from weeks 6–12 after initial treatment under commercial greenhouse conditions

| Treatment               | % Larvae      |               |               |               |
|-------------------------|---------------|---------------|---------------|---------------|
|                         | 6 weeks       | 8 weeks       | 10 weeks      | 12 weeks      |
| Control (marigold only) | 0.12 ± 0.06 a | 0.15 ± 0.08 a | 0.39 ± 0.15 a | 0.75 ± 0.27 a |
| Mites only              | 0.09 ± 0.04 a | 0.24 ± 0.20 a | 0.33 ± 0.10 a | 0.50 ± 0.14 b |
| ERL1170 + mites + fungi | 0.27 ± 0.12 a | 0.23 ± 0.10 a | 0.61 ± 0.24 a | 0.34 ± 0.10 a |
| GHA + mites + fungi     | 0.09 ± 0.04 a | 0.30 ± 0.15 a | 0.41 ± 0.18 a | 0.48 ± 0.23 a |
| $F_{3,120}$             | 1.08          | 0.244         | 0.418         | 0.562         |
| $P$                     | 0.361         | 0.865         | 0.740         | 0.641         |

Note: Means within columns with the same lowercase letters are not significantly different (Tukey's HSD,  $P > 0.05$ ).

mean number of WFT/crop plant was not significantly different relative to marigold treatment ( $F_{3,439} = 3.0$ ;  $P = 0.825$ ):  $0.37 \pm 0.08$  (ERL1170 + lure + mites),  $0.36 \pm 0.08$  (GHA + lure + mites),  $0.28 \pm 0.07$  (lure only) and  $0.35 \pm 0.08$  (untreated control) but the

number of WFT on crop plants in general increased significantly over time ( $F_{3,439} = 3.922$ ;  $P = 0.009$ ) (Table 5).

There were significant differences in the number of WFT on crop plants among greenhouses ( $F_{14,439} = 2.9$ ;  $P < 0.001$ ). This would be

expected because of the variation among houses in crops grown. For example, ornamental plants were inspected in all 15 greenhouses used in the analysis whereas in two greenhouse a mixture of herbs or vegetable starts in addition to ornamentals were sampled. Of the two houses containing herbs, the plants that were inspected in one were comprised of 63% that type. The majority of plants inspected (top 5) were in the *Calibrachoa*, *Ipomoea*, *Helichrysum*, *Begonia* and *Impatiens* genera, whereas 86% of the greenhouses contained two or more of those main varieties.

Damage on individual crop plants ranged from a rating of 0 (none) to 3 (26–50%) and increased over time as WFT numbers increased. There were no significant differences in crop plant damage associated with the marigold GPS treatments over time ( $F_{9,439} = 0.779$ ;  $P = 0.636$ ). In general, the overall mean damage rating on crop plants was low (0–10%) and not significantly different in respect to marigold treatment ( $F_{3,439} = 0.283$ ;  $P = 0.838$ ).

0.06 ± 0.02 (ERL1170 + lure + mites), 0.08 ± 0.03 (GHA + lure +

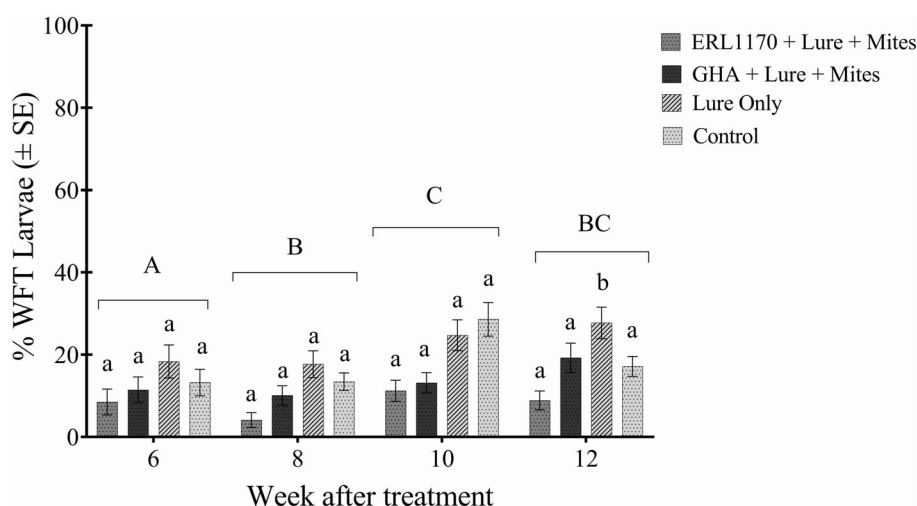
mites), 0.07 ± 0.02 (lure only) and 0.06 ± 0.02 (untreated control) and increased significantly over time ( $F_{3,439} = 5.909$ ;  $P = 0.001$ ).

### 3.2.4 Fungal persistence within marigold GPS

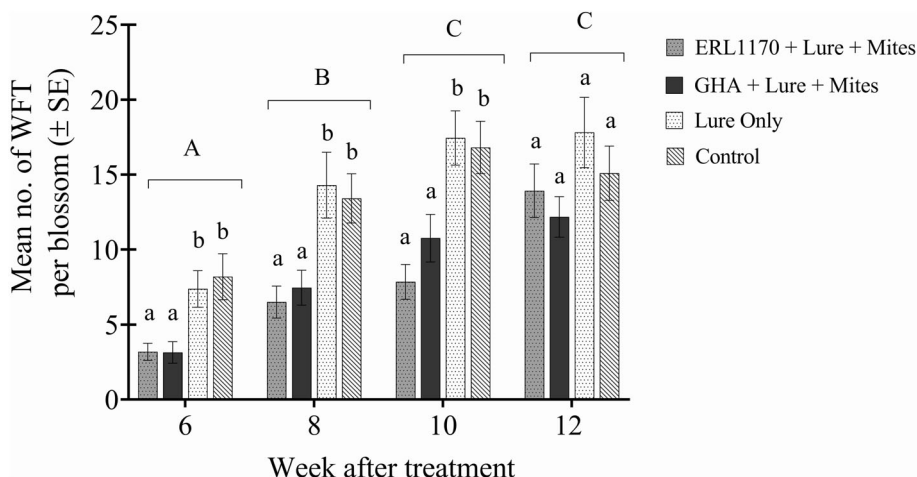
At 12 weeks after application of the granular fungal formulations, significant differences in the mean number of CFU g<sup>-1</sup> of soil were observed among GPS ( $F_{3,182} = 148.19$ ;  $P < 0.001$ ) (Fig. 4). A significantly higher concentration of CFU/per gram was obtained from the ERL1170 treatment ( $2.5 \times 10^5$  CFU g<sup>-1</sup> of soil) than the GHA ( $9.5 \times 10^4$  CFU g<sup>-1</sup> of soil) ( $P < 0.001$ ).

### 3.2.5 WFT and predatory mites within marigold blossoms

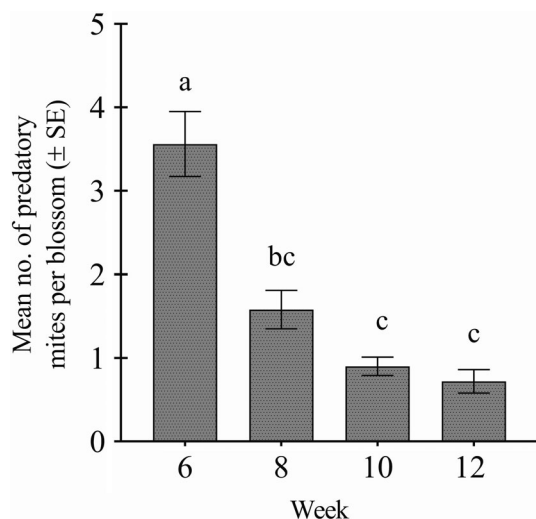
The mean number of WFT (adults + larvae) remaining within marigold blossoms were consistent with the WFT trends from plant tapping observed in the treatments over time ( $F_{9,231} = 0.517$ ;  $P = 0.862$ ) (Fig. 5). Marigolds treated with predatory mites, regardless of treatment with fungal granules, generally had significantly



**Figure 4.** Mean number of *Beauveria bassiana* colony forming units (CFU) (mean ± SE) from marigold GPS soil treated with granular fungal formulations of *Beauveria bassiana* experimental isolate ERL1170 and commercial isolate GHA, *Neoseiulus cucumeris* predatory mites within slow-release sachets and a pheromone lure and marigolds with and without the lure after 12 weeks under commercial greenhouse conditions. Means with the same lowercase letter are not significantly different among treatments (Tukey's HSD,  $P > 0.05$ ).



**Figure 5.** Mean number of *Frankliniella occidentalis* (WFT) adults and larvae per blossom from marigold GPS untreated or treated with granular fungal formulations of *Beauveria bassiana* experimental isolate ERL1170 or commercial isolate GHA in combination with *Neoseiulus cucumeris* predatory mites within slow-release sachets and a pheromone lure and marigolds with and without the lure from weeks 6–12 after initial treatment under commercial greenhouse conditions. Means with the same upper-case letters among weeks or the same lower-case letters among treatments within a week are not significantly different (Tukey's HSD,  $P > 0.05$ ).



**Figure 6.** Mean number of *Neoseiulus cucumeris* predatory mites per blossom from marigold GPS treated with the granular fungal formulations of *Beauveria bassiana* and a pheromone lure from weeks 6–12 after treatment within commercial greenhouses. Means with the same lowercase letters are not significantly different among weeks (Tukey's HSD,  $P > 0.05$ ).

fewer WFT per blossom than the untreated control ( $F_{3,77} = 8.77$ ;  $P < 0.001$ ). The mean number of WFT per blossom increased significantly from weeks 6–12 ( $F_{3,231} = 39.81$ ;  $P < 0.001$ ). The control marigolds with and without the lure consistently had significantly more WFT within their blossoms than those treated with fungi, mites and lures until week 12. Similar to the controlled greenhouse trials, larvae were the dominant life stage present within the blossoms after tapping (92%). A significant difference in the mean number of WFT larvae per blossom was observed among treatments ( $F_{3,77} = 9.29$ ;  $P < 0.001$ ) with the controls harboring significantly more larvae than the marigold GPS ( $P < 0.001$ ). No differences in WFT abundance were detected between the controls with and without the lure ( $P = 0.817$ ) or between the fungal granular + mite + lure treatments ( $P = 0.977$ ). For the adult stage, no differences were observed among treatments ( $F_{3,77} = 0.683$ ;  $P = 0.565$ ). Predatory mites were observed throughout the experimental period within marigold blossoms, but declined significantly from week 6 through 12 ( $F_{3,180} = 26.15$ ,  $P < 0.001$ ) (Fig. 6). There was no significant difference in the number of predatory mites in marigold blossoms between the different *B. bassiana* isolate treatments ( $F_{3,135} = 0.445$ ,  $P = 0.721$ ).

## 4 DISCUSSION

The overall results demonstrate that a marigold GPS can be an IPM strategy to suppress attracted WFT when used in greenhouse ornamental plant production. The predatory mite, *N. cucumeris*, played a major role in WFT suppression. Further development and testing of the system should be conducted to refine its effectiveness and to determine deployment costs for growers to adopt the strategy.

Several strains of *B. bassiana* are commercially available as different formulation types but no granular is available for use in the USA.<sup>51</sup> However, granular formulations of *B. bassiana* have been shown to be effective in suppressing WFT and other thrips species under laboratory and controlled greenhouse conditions by targeting the soil inhabiting stages.<sup>13,16,19,52–55</sup> The effectiveness of managing greenhouse pests using entomopathogenic

fungi depends on many factors including the strain/isolate, the formulation, the rate of application to maximize pest contact with infective conidia and microclimate.<sup>55,56</sup> Our previous work investigated the effectiveness of mycotized millet grains used alone, incorporated into the soil of marigold plants under controlled greenhouse cage experiments. Ninety percent fewer WFT were detected on plants treated with the experimental isolate ERL1170 than on the untreated plants.<sup>16</sup> We also found at 11 weeks post application, an average of  $3.0 \times 10^5$  CFU g<sup>-1</sup> of soil was obtained for ERL1170, which was significantly more than for the commercial GHA isolate ( $1.1 \times 10^4$  CFU g<sup>-1</sup>), a result like the fungal persistence observed in our commercial greenhouse trial (section 3.2.2). The results reported herein followed a similar trend as our previous work in that the number of WFT/plant for the control and GHA treatments were not significantly different at 8 weeks, resulting in reduced plant quality from WFT feeding by week 10. Similarly, from weeks 4–8, plants treated with the experimental isolate (ERL1170) had significantly fewer WFT than the untreated control and generally outperformed the commercial isolate (GHA). However, WFT suppression within the GPS system was not as effective in this study compared with our past research. Abiotic and biotic factors like temperature, relative humidity, sunlight, soil moisture and competition for nutrients by other soil saprophytes can affect the growth, stability and efficacy of entomopathogenic fungi.<sup>51,57</sup> It is possible there were microclimatic factors that reduced fungal infection of WFT in this trial. For example, a low moisture level in the pots linked with weather may have dried out the potting soil. It has been reported that the survival of *B. bassiana* conidia and growth from granules is significantly reduced at soil moisture levels of 20%.<sup>54</sup> Finally, attenuation (loss of viability, virulence and other traits) may also occur during storage, serial subculturing of isolates and quality of nutrition source which may have affected the performance of the isolates used in these trials compared to our previous trial.<sup>58–60</sup>

Successful IPM of WFT using a GPS that combines soil-based applications of a granular fungal formulation and predatory mites also depend on the timing of treatment relative to the life cycle and population dynamics of the pest.<sup>16</sup> The use of fungal granules alone to cause mortality to the pupal stage would take more time compared to if the adult/larval life stage were targeted when plants are first infested. This can be overcome, as demonstrated herein, by the addition of predatory mites. Predatory mites released from sachets persisted and contributed significantly to the reduction of WFT within the GPS blossoms and maintained larval populations at lower levels on foliage than untreated plants and those treated only with fungi. Mycoinsecticides are often used together with other biocontrol agents or as a backup when commercially reared natural enemies are not available or fail to adequately reduce pests.<sup>61,62</sup> In the GPS, the predatory mites act as the first line of defense against WFT. Ideally, larvae that escape predation are the target of fungi in the soil where most WFT pupate. Our marigold GPS results show WFT were suppressed primarily by the addition of mites and to a lesser extent by fungi. WFT suppression within the system was observed for up to 12 weeks under commercial settings with the replacement of mite sachets after 6 weeks per supplier guidelines and up to 10 weeks with one release under controlled conditions. Under the controlled conditions, replacement after 6 weeks would likely have provided a greater reduction in noticeable damage. Because we did not include fungi only treatments in the commercial greenhouse trial, it was not possible to distinguish treatment effects from fungi and mites separately.



Compatibility of natural enemies with other biocontrol agents to manage WFT is important for IPM success.<sup>7,61</sup> The foliar-dwelling predatory mites used in the GPS were applied using a slow-release packet hung in the marigold canopy whereas the fungus was applied as granules in the soil. This allowed for minimal mite contact with infective conidia. *Beauveria bassiana* has been reported to have little effect on *N. cucumeris* when contact with the mycoinsecticide occurs.<sup>21,23</sup> Predatory mites and entomopathogenic fungi can behave additively, synergistically or antagonistically.<sup>63</sup> Some studies reported *B. bassiana* had minimal effects on mortality of other predatory mites commonly used against WFT, including *Stratiolaelaps scimitus* (Womersley), formerly *Hypoaspis miles* (Berlese)<sup>19,22</sup> and *Neoseiulus* (*Amblyseius*) *barkeri* (Hughes).<sup>64,65</sup> However, reports have shown adverse effects like mortality from direct fungal applications to *Amblyseius swirskii* (Athias-Henriot),<sup>66</sup> indirect effects on the developmental time, longevity and fecundity of *N. barkeri* when fed *B. bassiana*-infected WFT,<sup>67</sup> and a reduction in their searching and predation rates.<sup>63</sup> The use of the marigold GPS in combination with other predatory mite species and fungal isolates should be investigated.

The deployment rate of GPS for optimal WFT management and GPS effectiveness over the growing season as crop plants start to bloom is unknown. The attractiveness of flowering marigolds to WFT has been observed to be higher early in the season when fewer crop plants are in bloom (C.F.S. unpublished data). This has also been observed when flowering chrysanthemums were used as trap plants.<sup>29</sup> Because WFT are weak fliers with limited dispersal capacity,<sup>68,69</sup> it is uncertain at what distance the GPS could attract WFT out of the crop. The GPS would probably have the greatest impact used to trap WFT from plants prone to infestations or near locations where WFT are known to be commonly observed within the greenhouses.

The pheromone lure used in this trial showed no evidence of additional WFT attraction. This could be due to the close deployment rate of the lures in combination with the GPS. Spacing of GPS in this study averaged 1/25 m<sup>2</sup> with no replacement of the lure over the 12 week period. Other studies have spaced lures 10 m apart to attract WFT to sticky cards<sup>44,45</sup> while the manufacturer recommendation is 1 lure/100 m<sup>2</sup> and to be replaced after 4–6 weeks (Biobest Group N.V. [https://www.plantproducts.com/ca/images/Thripher\\_lures\\_Biobest.pdf](https://www.plantproducts.com/ca/images/Thripher_lures_Biobest.pdf)). Our results suggested that marigolds were naturally attractive. However, WFT were found first on crop plants close to marigold GPS with lures rather than those without lures. Further investigations using the GPS should focus on deployment rates with and without lures, the effect of initial WFT infestation level on level of suppression and effect of different rates and timing of predatory mite and/or granular formulation applications to improve efficacy.

## 5 CONCLUSION

The growing season for greenhouse ornamentals in the Northeast USA occurs typically from March to June. These trials demonstrated marigolds attracted WFT in commercial greenhouses containing a variety of ornamentals and vegetable/herb starts. Results suggest the marigold GPS has the potential to provide WFT suppression within the system for 6–12 weeks when mites are applied according to manufacturer recommendations. The foliar-dwelling predatory mites acted as the first line of defense to provide suppression which prolonged the use of the marigold. Infection from the granular fungal formulation in the soil played a minor role in WFT suppression within this system. A granular

formulation of *B. bassiana* is not currently available in the USA and refinement of the granular formulation is necessary within the marigold GPS. However, growers could still use predatory mite sachets to reduce WFT populations. Crop monitoring is a cornerstone of IPM and careful monitoring of these systems is essential to ensure WFT do not build up and act as a source of WFT to the crops. Grower adoption of a marigold GPS would provide a valuable way for them to trap, monitor and manage WFT early in the crop production cycle before other plants are in bloom and damaging infestation levels become evident.

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## DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

## SUPPORTING INFORMATION

Supporting information may be found in the online version of this article.

## REFERENCES

- 1 Cloyd RA, Western flower thrips (*Frankliniella occidentalis*) management on ornamental crops grown in greenhouses. Have we reached an impasse? *Pest Technol* **3**:1–9 (2009).
- 2 Kirk WD and Terry LI, The spread of the western flower thrips, *Frankliniella occidentalis* (Pergande). *Agric Forest Entomol* **5**:301–310 (2003). <https://doi.org/10.1046/j.1461-9563.2003.00192.x>.
- 3 Reitz SR, Gao Y, WDJ K, Hoddle MS, Leiss KA and Funderburk JE, Invasion biology, ecology, and management of western flower thrips. *Annu Rev Entomol* **65**:17–37 (2020). <https://doi.org/10.1146/annurev-ento-011019-024947>.
- 4 Bielza P, Insecticide resistance management strategies against the western flower thrips, *Frankliniella occidentalis*. *Pest Manag Sci* **64**: 1131–1138 (2008). <https://doi.org/10.1002/ps.1620>.
- 5 Cloyd RA, Western flower thrips (Thysanoptera: Thripidae) and insecticide resistance: an overview and strategies to mitigate insecticide resistance development. *J Entomol Sci* **51**:257–273 (2016). <https://doi.org/10.18474/JES16-15.1>.
- 6 Gao Y, Lei Z and Reitz SR, Western flower thrips resistance to insecticides: detection, mechanisms and management strategies. *Pest Manag Sci* **68**:1111–1121 (2012). <https://doi.org/10.1002/ps.3305>.
- 7 Mouden S, Sarmiento KF, Klinkhamer PG and Leiss KA, Integrated pest management in western flower thrips: past, present and future. *Pest Manag Sci* **73**:813–822 (2017). <https://doi.org/10.1002/ps.4531>.
- 8 Li J, Gómez M, Rickard B and Skinner M, Factors influencing adoption of integrated pest management in Northeast greenhouse and nursery

- production. *Agric Resour Econ Rev* **42**:310–324 (2013). <https://doi.org/10.1017/S1068280500004391>.
- 9 van Lenteren JC and Nicot PC, Integrated pest management methods and considerations concerning implementation in greenhouses, in *Integrated Pest and Disease Management in Greenhouse Crops, Plant Pathology in the 21st Century 9*, ed. by Gullino M, Albajes R and Nicot P. Cham, Springer, pp. 177–193 (2020).
- 10 Barzman M, Bärberi P, Birch AN, Boonekamp PG, Dachbrodt-Saaydeh S, Graf B *et al.*, Eight principles of integrated pest management. *Agron Sustain Dev* **35**:1199–1215 (2015). <https://doi.org/10.1007/s13593-015-0327-9>.
- 11 Skinner M, Sullivan CF and Parker BL, Chapter 15: integrated pest management in greenhouse and other protected environments, in *Integrated Management of Insect Pests: Current and Future Developments*, ed. by Kogan M and Heinrichs EA. Burleigh Dodds Science Publishers, Cambridge (2019).
- 12 Ansari MA, Brownbridge M, Shah FA and Butt TM, Efficacy of entomopathogenic fungi against soil-dwelling life stages of western flower thrips, *Frankliniella occidentalis*, in plant-growing media. *Entomol Exp Appl* **127**:80–87 (2008). <https://doi.org/10.1111/j.1570-7458.2008.00674.x>.
- 13 Davari A, Parker B, Sullivan C, Ghalegholabbehbahani A and Skinner M, Biological control of Western flower thrips, *Frankliniella occidentalis* using a self-sustaining granular fungal treatment. *Bull Entomol Res* **111**:688–693 (2021). <https://doi.org/10.1017/S0007485321000365>.
- 14 Gerson U and Weintraub PG, Mites (Acari) as a factor in greenhouse management. *Annu Rev Entomol* **57**:229–247 (2011). <https://doi.org/10.1146/annurev-ento-120710-100639>.
- 15 Hewitt LC, Shipp L, Buitenhuis R and Scott-Dupree C, Seasonal climatic variations influence the efficacy of predatory mites used for control of western flower thrips in greenhouse ornamental crops. *Exp Appl Acarol* **65**:435–450 (2015). <https://doi.org/10.1007/s10493-014-9861-4>.
- 16 Skinner M, Gouli S, Frank CE, Parker BL and Kim JS, Management of *Frankliniella occidentalis* (Thysanoptera: Thripidae) with granular formulations of entomopathogenic fungi. *Biol Control* **63**:246–252 (2012). <https://doi.org/10.1016/j.biocontrol.2012.08.004>.
- 17 Van Driesche RG, Lyon S, Stanek EJ III, Xu B and Nunn C, Evaluation of efficacy of *Neoseiulus cucumeris* for control of western flower thrips in spring bedding crops. *Biol Control* **36**:203–215 (2006). <https://doi.org/10.1016/j.BIOCONTROL.2005.08.011>.
- 18 Zhang T, Reitz SR, Wang H and Lei Z, Sublethal effects of *Beauveria bassiana* (Ascomycota: Hypocreales) on life table parameters of *Frankliniella occidentalis* (Thysanoptera: Thripidae). *J Econ Entomol* **108**:975–985 (2015). <https://doi.org/10.1093/jeetov091>.
- 19 Zhang X, Wu S, Reitz SR and Gao Y, Simultaneous application of entomopathogenic *Beauveria bassiana* granules and predatory mites *Stratiolaelaps scimitus* for control of western flower thrips, *Frankliniella occidentalis*. *J Pest Sci* **94**:119–127 (2020). <https://doi.org/10.1007/s10340-020-01227-5>.
- 20 Saito T and Brownbridge M, Compatibility of soil-dwelling predators and microbial agents and their efficacy in controlling soil dwelling stages of western flower thrips, *Frankliniella occidentalis*. *Biol Control* **92**:92–100 (2016). <https://doi.org/10.1016/j.biocontrol.2015.10.003>.
- 21 Saito T and Brownbridge M, Compatibility of foliage-dwelling predatory mites and mycoinsecticides, and their combined efficacy against western flower thrips *Frankliniella occidentalis*. *J Pest Sci* **91**:1–10 (2018). <https://doi.org/10.1007/s10340-018-0991-z>.
- 22 Sun W, Sarkar SC, Xu X, Lei Z, Wu S and Meng R, The entomopathogenic fungus *Beauveria bassiana* used as granules has no impact on the soil-dwelling predatory mite *Stratiolaelaps scimitus*. *Syst Appl Acarol* **23**:2165–2172 (2018). <https://doi.org/10.11158/saa.23.11.9>.
- 23 Jacobson RJ, Chandler D, Fenlon J and Russell KM, Compatibility of *Beauveria bassiana* (Balsamo) Vuillemin with *Amblyseius cucumeris* Oudemans (Acarina: Phytoseiidae) to control *Frankliniella occidentalis* Pergande (Thysanoptera: Thripidae) on cucumber plants. *Biocontrol Sci Technol* **11**:391–400 (2001). <https://doi.org/10.1080/09583150120055808>.
- 24 Bennison J, Maulden K, Dewhurst S, Pow E, Slatter P and Wadhams L, Towards the development of a push-pull strategy for improving biological control of western flower thrips on chrysanthemum, in *Thrips and Tospoviruses: Proceedings of the 7th International Symposium on Thysanoptera*, ed. by Mound LA and Marullo R. Australian National Insect Collection, Canberra, pp. 199–206 (2002).
- 25 Bhattacharyya M, The push-pull strategy: a new approach to the eco-friendly method of pest management in agriculture. *J Entomol Zool Stud* **5**:604–607 (2017).
- 26 Cook SM, Khan ZR and Pickett JA, The use of push-pull strategies in integrated pest management. *Annu Rev Entomol* **52**:375–400 (2007). <https://doi.org/10.1146/annurev.ento.52.110405.091407>.
- 27 Parolin P, Bresch C, Desneux N, Brun R, Bout A, Boll R *et al.*, Secondary plants used in biological control: a review. *Int J Pest Manage* **58**:91–100 (2012). <https://doi.org/10.1080/09670874.2012.659229>.
- 28 Berlinger MJ, Lok-Van Dijk B, Dahan R, Lebiush-Mordechai S and RAJ T, Indicator plants for monitoring pest population growth. *Ann Entomol Soc Am* **89**:611–622 (1996). <https://doi.org/10.1093/aesa/89.5.611>.
- 29 Buitenhuis R and Shipp JL, Factors influencing the use of trap plants for the control of *Frankliniella occidentalis* (Thysanoptera: Thripidae) on greenhouse potted chrysanthemum. *Environ Entomol* **35**:1411–1416 (2006). <https://doi.org/10.1093/ee/35.5.1411>.
- 30 Shelton AM and Badenes-Perez FR, Concepts and applications of trap cropping in pest management. *Annu Rev Entomol* **51**:285–308 (2006). <https://doi.org/10.1146/annurev.ento.51.110104.150959>.
- 31 Sarkar SC, Wang E, Wu S and Lei Z, Application of trap cropping as companion plants for the management of agricultural pests: a review. *Insects* **9**:128 (2018). <https://doi.org/10.3390/insects9040128>.
- 32 Frank SD, Biological control of arthropod pests using banker plant systems: past progress and future directions. *Biol Control* **52**:8–16 (2010). <https://doi.org/10.1016/j.BIOCONTROL.2009.09.011>.
- 33 Huang N, Enkegaard A, Osborne LS, Ramakers PMJ, Messelink GJ, Pijnakker J *et al.*, The banker plant method in biological control. *Crit Rev Plant Sci* **30**:259–278 (2011). <https://doi.org/10.1080/07352689.2011.572055>.
- 34 Osborne LS, Landa Z, Taylor DJ and Tyson RV, Using banker plants to control insects in greenhouse vegetables. *Proc Fla State Hort Soc* **118**:127–128 (2005).
- 35 Waite MO, Scott-Dupree CD, Brownbridge M, Buitenhuis R and Murphy G, Evaluation of seven plant species/cultivars for their suitability as banker plants for *Orius insidiosus* (Say). *BioControl* **59**:79–87 (2014). <https://doi.org/10.1007/s10526-013-9549-4>.
- 36 Glenister CS, Skinner M and Frank C, Guardian plants support greenhouse IPM as indicators, traps, bankers and habitat plants. Entomological Society of America Annual Meeting; Nov. 16–19; Reno, NV (2008).
- 37 O'Connell D, Using plants to fight pests. *Grower Talks Features*, 7 July 2017. <https://www.growertalks.com/Article/?articleID=15844> (Accessed 22 February 2022).
- 38 Quarles W, Fatal attraction-guardians, bankers, trap crops and greenhouse biocontrol. *IPM Pract* **32**:1–8 (2010).
- 39 Blumthal MR, Cloyd RA, Spomer LA and Warnock DF, Flower color preferences of western flower thrips. *HortTechnology* **15**:846–853 (2005).
- 40 Gillespie DR and Vernon RS, Trap catch of western flower thrips (Thysanoptera: Thripidae) as affected by color and height of sticky traps in mature greenhouse cucumber crops. *J Econ Entomol* **83**:971–975 (1990). <https://doi.org/10.1093/jeet/83.3.971>.
- 41 Pizzol J, Nammour D, Hervouet JP, Bout A, Desneux N and Maillet L, Comparison of two methods of monitoring thrips populations in a greenhouse rose crop. *J Pest Sci* **83**:191–196 (2010). <https://doi.org/10.1007/s10340-010-0286-5>.
- 42 Peres FSC, Fernandes AO, Silveira LCP and Silva CSB, Marigold as attractive plant for thrips in protected organic melon cultivation. *Bragantia* **68**:953–960 (2009). <https://doi.org/10.1590/S0006-87052009000400015>.
- 43 Buitenhuis R, Shipp JL, Jandricic S, Murphy G and Short M, Effectiveness of insecticide-treated and non-treated trap plants for the management of *Frankliniella occidentalis* (Thysanoptera: Thripidae) in greenhouse ornamentals. *Pest Manag Sci* **63**:910–917 (2007). <https://doi.org/10.1002/ps.1426>.
- 44 Abdullah ZS, Greenfield BP, Ficken KJ, Taylor JW, Wood M and Butt TM, A new attractant for monitoring western flower thrips, *Frankliniella occidentalis* in protected crops. *Springerplus* **4**:89 (2015). <https://doi.org/10.1186/s40064-015-0864-3>.
- 45 Broughton S and Harrison J, Evaluation of monitoring methods for thrips and the effect of trap colour and semiochemicals on sticky trap capture of thrips (Thysanoptera) and beneficial insects (Syrphidae, Hemerobiidae) indeciduous fruit trees in Western Australia. *Crop Prot* **42**:156–163 (2012). <https://doi.org/10.1016/j.cropro.2012.05.004>.

- 46 Tillman PG and Cottrell TE, Case study: trap crop with pheromone traps for suppressing *Euschistus servus* (Heteroptera: Pentatomidae) in cotton. *Psyche: J Entomol* **2012**:1–10 (2012). <https://doi.org/10.1155/2012/401703>.
- 47 Broughton S, Cousins DA and Rahman T, Evaluation of semiochemicals for their potential application in mass trapping of *Frankliniella occidentalis* (Pergande) in roses. *Crop Prot* **67**:130–135 (2015). <https://doi.org/10.1016/j.cropro.2014.10.011>.
- 48 Gouli S, Gouli V, Skinner M, Parker BL, Marcelino J and Shternshis M, Mortality of western flower thrips, *Frankliniella occidentalis*, under influence of single and mixed fungal inoculations. *J Agric Technol* **4**: 37–47 (2008).
- 49 Hywel-Jones NL and Gillespie AT, Effect of temperature on spore germination in *Metarhizium anisopliae* and *Beauveria bassiana*. *Mycol Res* **94**:389–392 (1990). [https://doi.org/10.1016/S0953-7562\(09\)80363-8](https://doi.org/10.1016/S0953-7562(09)80363-8).
- 50 Chase A, Osborne L and Ferguson V, Selective isolation of the entomopathogenic fungi *Beauveria bassiana* and *Metarhizium anisopliae* from an artificial potting medium. *Fla Entomol* **69**:285–292 (1986). <https://doi.org/10.2307/3494930>.
- 51 Mascarin GM and Jaronski ST, The production and uses of *Beauveria bassiana* as a microbial insecticide. *World J Microbiol Biotechnol* **32**: 177 (2016). <https://doi.org/10.1007/s11274-016-2131-3>.
- 52 Lee SJ, Kim S, Kim JC, Lee MR, Hossain MS, Shin TS et al., Entomopathogenic *Beauveria bassiana* granules to control soil-dwelling stage of western flower thrips, *Frankliniella occidentalis* (Thysanoptera: Thripidae). *BioControl* **62**:639–648 (2017). <https://doi.org/10.1007/s10526-017-9818-8>.
- 53 Li D, Park SE, Lee MR, Kim JC, Lee SJ and Kim JS, Soil application of *Beauveria bassiana* JEF-350 granules to control melon thrips, *Thrips palmi* Karny (Thysanoptera: Thripidae). *J Asia Pac Entomol* **24**:636–644 (2021). <https://doi.org/10.1016/J.ASPEN.2021.05.010>.
- 54 Zhang XR, Lei ZR, Reitz SR, Wu SY and Gao YL, Laboratory and greenhouse evaluation of a granular formulation of *Beauveria bassiana* for control of western flower thrips, *Frankliniella occidentalis*. *Insects* **10**:58 (2019). <https://doi.org/10.3390/insects10020058>.
- 55 Shipp JL, Zhang Y, Hunt DWA and Ferguson G, Influence of humidity and greenhouse microclimate on the efficacy of *Beauveria bassiana* (Balsamo) for control of greenhouse arthropod pests. *Environ Entomol* **32**:1154–1163 (2003). <https://doi.org/10.1603/0046-225X-32.5.1154>.
- 56 Ugine TA, Wraight SP and Sanderson JP, Effects of manipulating spray-application parameters on efficacy of the entomopathogenic fungus *Beauveria bassiana* against western flower thrips, *Frankliniella occidentalis*, infesting greenhouse impatiens crops. *Biocontrol Sci Technol* **17**:193–219 (2007). <https://doi.org/10.1080/09583150600937618>.
- 57 Lingg AJ and Donaldson MD, Biotic and abiotic factors affecting stability of *Beauveria bassiana* conidia in soil. *J Invertebr Pathol* **38**:191–200 (1981). [https://doi.org/10.1016/0022-2011\(81\)90122-1](https://doi.org/10.1016/0022-2011(81)90122-1).
- 58 Safavi SA, Attenuation of the entomopathogenic fungus *Beauveria bassiana* following serial in vitro transfers. *Biologia* **67**:1062–1068 (2012). <https://doi.org/10.2478/s11756-012-0120-z>.
- 59 Safavi SA, Shah FA, Pakdel AK, Rasoulzadeh GR, Bandani AR and Butt TM, Effect of nutrition on growth and virulence of the entomopathogenic fungus *Beauveria bassiana*. *FEMS Microbiol Lett* **270**:116–123 (2007). <https://doi.org/10.1111/j.1574-6968.2007.00666.x>.
- 60 Sandhu SS, Rajak RC and Agarwal GP, Studies on prolonged storage of *Beauveria bassiana* conidia: effects of temperature and relative humidity on conidial viability and virulence against chickpea borer, *Helicoverpa armigera*. *Biocontrol Sci Technol* **3**:47–53 (1993). <https://doi.org/10.1080/09583159309355258>.
- 61 Gonzalez F, Tkaczuk C, Dinu MM, Fiedler Z, Vidal S, Zchori-Fein E et al., New opportunities for the integration of microorganisms into biological pest control systems in greenhouse crops. *J Pest Sci* **89**: 295–311 (2016). <https://doi.org/10.1007/s10340-016-0751-x>.
- 62 van Lenteren JC, Alomar O, Ravensberg WJ and Urbaneja A, Biological control agents for control of pests in greenhouses, in *Integrated Pest and Disease Management in Greenhouse Crops, Plant Pathology in the 21st Century* 9, ed. by Gullino M, Albajes R and Nicot P. Cham, Springer, pp. 409–493 (2020).
- 63 Wu S, Gao Y, Smagghe G, Xu X and Lei Z, Interactions between the entomopathogenic fungus *Beauveria bassiana* and the predatory mite *Neoseiulus barkeri* and biological control of their shared prey/host *Frankliniella occidentalis*. *Biol Control* **98**:43–51 (2016). <https://doi.org/10.1016/J.BIOCONTROL.2016.04.001>.
- 64 Wu SY, Gao YL, Zhang YP, Wang ED, Xu XN and Lei ZR, An entomopathogenic strain of *Beauveria bassiana* against *Frankliniella occidentalis* with no detrimental effect on the predatory mite *Neoseiulus barkeri*: evidence from laboratory bioassay and scanning electron microscopic observation. *PLoS One* **9**:e84732 (2014). <https://doi.org/10.1371/journal.pone.0084732>.
- 65 Wu SY, Gao YL, Xu XN, Goettel MS and Lei ZR, Compatibility of *Beauveria bassiana* with *Neoseiulus barkeri* for control of *Frankliniella occidentalis*. *J Integr Agric* **14**:98–105 (2015). [https://doi.org/10.1016/S2095-3119\(13\)60731-5](https://doi.org/10.1016/S2095-3119(13)60731-5).
- 66 Seiedy M, Tork M and Deyhim F, Effect of the entomopathogenic fungus *Beauveria bassiana* on the predatory mite *Amblyseius swirskii* (Acari: Phytoseiidae) as a non-target organism. *Syst Appl Acarol* **20**: 241–250 (2015). <https://doi.org/10.11158/saa.20.3.2>.
- 67 Wu S, Gao Y, Xu X, Wang D, Li J, Wang H et al., Feeding on *Beauveria bassiana*-treated *Frankliniella occidentalis* causes negative effects on the predatory mite *Neoseiulus barkeri*. *Sci Rep* **5**:12033 (2015). <https://doi.org/10.1038/srep12033>.
- 68 Hollister B, Cameron EA and Teulon DAJ, Effect of p-anisaldehyde and a yellow color on behavior and capture of western flower thrips, in *Thrips Biology and Management*, ed. by Parker BL, Skinner M and Lewis T. Plenum Press, New York, pp. 571–574 (1995).
- 69 Rhoads M and Shipp L, Dispersal of adult western flower thrips (Thysanoptera: Thripidae) on chrysanthemum plants: impact of feeding-induced senescence of inflorescences. *Environ Entomol* **32**:1056–1065 (2003). <https://doi.org/10.1603/0046-225X-32.5.1056>.